Yen-Hsiang (Robert) Huang

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Educations

Ph.D. in Electrical and Computer Engineering | GPA: 4.0/4.0 Sept. 2022 - Present
Georgia Institute of Technology (Georgia Tech), Atlanta, GA, USA
B.S. in Electrical Engineering | GPA: 4.06/4.30 Sept. 2017 - July 2022
National Taiwan University (NTU), Taipei, Taiwan

Research Experience

Graduate Research Assistant
SCCAD (former GTCAD), Georgia Institute of Technology, Advisor: Prof. Sung Kyu Lim Sept. 2022 - Present

3D IC Optimization Fall 2024 - Present

- Timing optimization through reduced critical-path snaking and inter-die cell relocation (Snake-3D).
- Developed an RC-calibrated differentiable clock-tree optimization framework for 3D ICs, incorporating interconnect RC effects into timing-aware clock-tree optimization (CLK-3D).

Semiconductor Research Corporation (SRC) – JUMP 2.0 CHIMES Fall 2023 - Spring 2024

- Bridged the PPA gap between legacy-node 3D ICs and advanced-node 2D ICs.
- Reduced WNS of legacy-node 3D ICs by 50–93% compared to SOTA, achieving 87.9–97.7% of advanced-node 2D performance.

Die Bonding Bumps and Pads Legalization for 3D ICs Sept. 2022 - May 2023

- Developed a 3D-via legalization stage with a refinement step while minimizing runtime impact; see the related TCAD journal article and ISPD conference paper.
- Eliminated all 3D-via overlaps when via utilization is less than 40%.

Individual Study
EDA Lab., National Taiwan University, Advisor: Prof. Yao-Wen Chang Apr. 2020 - Sept. 2021

Independent Study on Physical Design

- Participated in physical-design-related global contests three times across different teams and placed in the top five twice (ISPD contest 2021 and ICCAD contest 2021).

Individual Study
ALCom Lab., National Taiwan University, Advisor: Prof. Jie-Hong Jiang Sept. 2020 - June 2021

Independent Study on Reversible Circuit Synthesis

- Implemented a reversible circuit synthesizer for controlled-NOT (CNOT) gates.

Work Experience

R&D Engineer Intern **Synopsys Headquarters**
Synopsys Inc. May 2025 - Dec. 2025

- Investigated machine-learning-based signal-integrity (SI) analysis for 2.5D ICs, evaluating multiple models to predict final-stage SI performance from early-stage design data.
- Independently implemented two provided prototypes in C++ and merged the resulting code into the main branch.

Software Engineer R&D Intern — **Hsinchu, Taiwan**

Synopsys Inc. – Concurrent Clock-and-Data (CCD) Team — July 2021 - Aug. 2021

- Enhanced the flip-flop merging algorithm in Synopsys IC Compiler II (ICC II) by considering flip-flop timing (slack).
- Improved timing quality in 13 real industrial benchmarks: 5.14%/3.47% in worst/total negative slack.

Software Engineer R&D Intern — **Hsinchu, Taiwan**

Cadence Inc. – Conformal Team — June 2020 - Sept. 2020

- Provided electronic change order (ECO) test cases in a five-person team for Cadence Conformal's robustness test.
- Generated 2,000+ cases across five categories in which Conformal acted unusually and proposed three improved flows to address them.

Publications

- **Yen-Hsiang Huang** and Sung Kyu Lim, "CLK-3D: RC-Calibrated Differentiable Clock-Tree Optimization for 3D ICs," IEEE/ACM International Conference on Computer-Aided Design, 2026.
- **Yen-Hsiang Huang** and Sung Kyu Lim, "Snake-3D: Differentiable Learning for Cross-Tier Logic Path Snaking Optimization in 3D ICs," IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2025. [PDF]
- **Yen-Hsiang Huang**, Sai Pentapati, Anthony Agnesina, Moritz Brunion, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024. [PDF]
- Sai Pentapati, Anthony Agnesina, Moritz Brunion, **Yen-Hsiang Huang**, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," ACM International Symposium on Physical Design (ISPD), 2023. [PDF]

Selected Projects

Mixed-Size Macro Placement and Legalization — Feb. 2021 - July 2021

- Developed a mixed-size macro placer for non-rectangular designs within a two-person team, maximizing the area for standard cells while maintaining the minimum channel needed by macros.

Honors and Awards

Ministry of Science and Technology Undergraduate Research Grant — July 2021

- Topic: Wafer-Scale Physics Modeling for Finite Element Methods.

Honorable Mention Award in the ACM ISPD contest 2021 — March 2021

- 4th place worldwide in the contest.

TSMC Scholarship for Undergraduate Students — Sept. 2020 - Jan. 2021

- Awarded to approximately 20 undergraduate students in Taiwan every semester.

The Dean's List Award — Spring 2020

- Ranked 1st, with a 4.30/4.30 GPA in the semester.

4th in Data Structure and Programming Contest — March 2019

- Cadence Inc. sponsored the contest, with 125 contestants.

Skills

Programming Languages

- C/C++ (expert), Python, MATLAB, LaTeX.

Languages

- English, Mandarin (native).